

**EXAMINING LIFESTYLE FACTORS
AND METABOLIC HEALTH MARKERS
ASSOCIATED WITH DIABETES USING
NHANES DATA (2021–2023)**

A Research Project – I Report

Submitted in partial fulfilment of the requirements for the
Award of the degree of

Master of Computer Applications

By

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DECLARATION BY THE STUDENT

To whom-so-ever it may concern

I, **Pargat Singh, 322201210**, hereby declare that the work done by me on **"Examining Lifestyle Factors and Metabolic Health Markers Associated with Diabetes Using NHANES Data (2021-2023)"**, is a record of original work for the partial fulfilment of the requirements for the award of the degree, Master of Computer Applications.

Pargat Singh (322201210)

Signature of the student: 

Dated: 20/01/2026

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I am deeply grateful to the Centers for Disease Control and Prevention (CDC) for making the NHANES data freely available for research purposes. This comprehensive dataset has provided the foundation for exploring important questions about diabetes and lifestyle factors in the American population.

I would also like to thank my family and friends for their constant encouragement and understanding throughout this academic journey. Their support has been a source of strength and motivation.

ABSTRACT

Diabetes mellitus is one of the fastest-growing chronic diseases worldwide, affecting over 38 million Americans. Understanding the relationship between lifestyle factors and diabetes prevalence is essential for developing effective prevention strategies. This Research Project–I establishes the foundation for analyzing this relationship using current health surveillance data.

The study uses the National Health and Nutrition Examination Survey (NHANES) 2011–2018 dataset from the Centers for Disease Control and Prevention. This dataset provides comprehensive health information including demographic characteristics, metabolic health indicators such as BMI, fasting glucose, and HbA1c levels, and lifestyle factors including sedentary behavior, smoking status, and dietary intake.

This report accomplishes four main objectives. First, it reviews existing literature to define the research problem. Second, it identifies gaps in current knowledge about diabetes and lifestyle factors. Third, it formulates seven specific research questions addressing prevalence patterns, lifestyle associations, and health disparities. Fourth, it outlines the analytical methodology using R programming for statistical analysis.

The expected outcome is a well-defined research framework ready for implementation in Research Project–II. The findings will help identify which lifestyle and demographic factors show the strongest associations with diabetes, potentially informing prevention programs and health education strategies.

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LIST OF ABBREVIATIONS

BMI - Body Mass Index

CDC - Centers for Disease Control and Prevention

DXA - Dual-Energy X-ray Absorptiometry

HbA1c - Glycated Hemoglobin (Hemoglobin A1c)

NHANES - National Health and Nutrition Examination Survey

R - Statistical Programming Language

T2DM - Type 2 Diabetes Mellitus

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CHAPTER 1: INTRODUCTION

1.1 Background and Context

Diabetes is one of the most pressing health challenges facing the world today. It is a chronic metabolic disorder characterized by elevated blood glucose levels due to insufficient insulin production or improper insulin utilization by the body. This condition affects millions of people globally and creates serious health complications including cardiovascular disease, kidney failure, nerve damage, and vision loss.

In the United States, the diabetes epidemic has reached alarming proportions. More than 38 million Americans are currently living with diabetes, representing approximately 11.6 percent of the population². Perhaps even more concerning is that nearly one in five people with diabetes remain undiagnosed, missing crucial opportunities for early intervention and treatment². The economic burden is staggering, with diabetes-related healthcare costs exceeding \$327 billion annually.

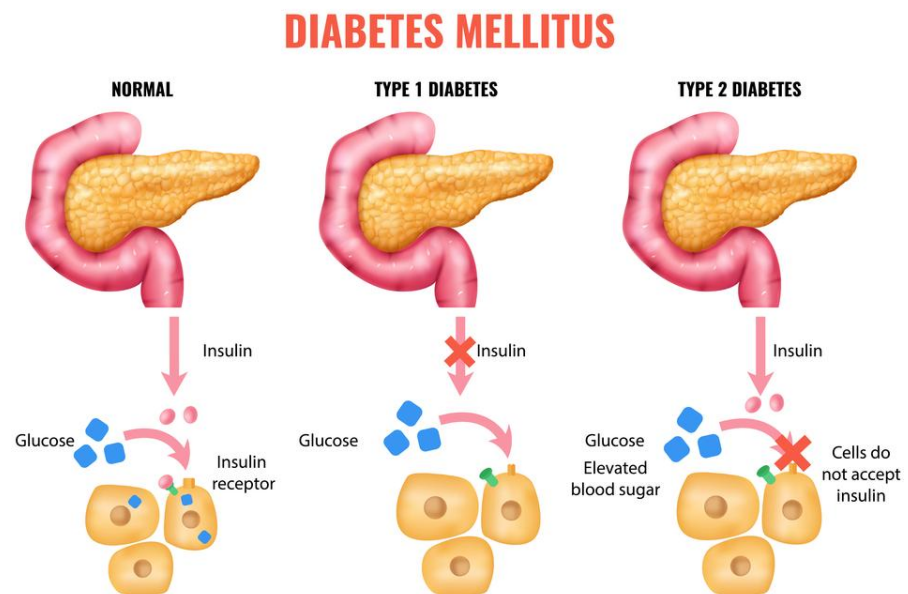


Figure 1.1: Mechanisms of Type 1 and Type 2 Diabetes⁶

Type 2 diabetes mellitus accounts for approximately 90 to 95 percent of all diabetes cases. Unlike Type 1 diabetes, which is typically diagnosed

in childhood and involves autoimmune destruction of insulin-producing cells, Type 2 diabetes develops gradually over time and is strongly influenced by lifestyle factors. This distinction is critical because it means Type 2 diabetes is largely preventable through modifications in diet, physical activity, and weight management.

Understanding how lifestyle choices affect diabetes risk is fundamental to prevention efforts. Factors such as sedentary behavior, dietary patterns, smoking status, and body weight all play important roles in determining who develops diabetes and who does not. However, the relative importance of these factors and their interactions in contemporary populations requires ongoing investigation using current data.

1.2 Importance of Data Science in Health Research

Modern healthcare generates vast amounts of data from surveys, electronic health records, laboratory tests, and epidemiological studies. However, raw data alone provides limited value. The real insights emerge through systematic analysis using statistical and computational methods.

Data science bridges the gap between data collection and actionable knowledge. It combines statistical analysis, programming skills, and domain expertise to extract meaningful patterns from complex datasets. For Master of Computer Applications students, applying these skills to real-world health problems demonstrates the practical value of computational methods beyond traditional software development.

This project utilizes R programming language, which has become the standard tool for statistical analysis in research and industry. R provides comprehensive libraries for data manipulation, statistical testing, and visualization⁵. Learning to work with real epidemiological data develops practical skills that are highly valued in healthcare analytics, public health institutions, pharmaceutical companies, and health technology organizations.

1.3 About the NHANES Dataset

The National Health and Nutrition Examination Survey (NHANES) is a cornerstone of public health surveillance in the United States. Conducted by the Centers for Disease Control and Prevention, NHANES has been collecting health information about Americans since the 1960s. The survey became a continuous program in 1999, releasing new data in two-year cycles.

What distinguishes NHANES from simple questionnaire surveys is its comprehensive approach. Participants undergo complete physical examinations, laboratory testing, and detailed interviews. This multi-modal data collection provides objective measurements alongside self-reported information. The survey captures demographics, medical history, dietary intake, physical activity patterns, body measurements, blood tests, and numerous other health indicators.

For this project, I am analyzing data from the 2021-2023 cycle, which represents the most recent complete dataset available². This timeframe is particularly relevant as it captures health behaviors and outcomes during the post-COVID-19 pandemic period, when many Americans experienced significant lifestyle changes. The dataset includes information from thousands of participants selected through a complex probability sampling design to ensure national representativeness.

1.4 Significance of This Research

This research addresses several important needs in diabetes epidemiology and prevention science. First, diabetes prevalence continues to rise despite decades of public health efforts¹. Understanding current patterns and risk factors is essential for developing more effective interventions. Using the most recent data available ensures our findings reflect contemporary health behaviors rather than outdated patterns.

Second, lifestyle habits have evolved significantly in recent years. The COVID-19 pandemic accelerated remote work adoption, potentially

increasing sedentary behavior. Food delivery services have changed dietary patterns. These shifts necessitate updated analyses of how contemporary lifestyle factors relate to diabetes risk.

Third, this project demonstrates technical competence in data science methodologies. It showcases abilities in data acquisition, data cleaning, statistical analysis, hypothesis testing, and result interpretation. These are core competencies for any data-focused career in healthcare or technology sectors.

Finally, the research addresses health equity concerns. Diabetes disproportionately affects certain racial and ethnic minority populations². By examining these disparities in current data, we can identify groups that would benefit most from targeted prevention programs and better understand potential barriers to diabetes prevention in vulnerable communities.

1.5 Research Questions

This study is guided by seven specific research questions that systematically examine different aspects of diabetes prevalence and lifestyle factors:

- What is the prevalence of diabetes in the NHANES 2021-2023 sample, and how does it vary by age, gender, and race/ethnicity?
- Is there a significant relationship between sedentary behavior (minutes per day) and diabetes status?
- Does current smoking status relate to diabetes prevalence?
- Is daily carbohydrate intake significantly different between individuals with and without diabetes?
- How do metabolic markers (BMI, fasting glucose, HbA1c) compare between diabetes and non-diabetes groups?
- Are there significant racial and ethnic disparities in diabetes prevalence and metabolic health indicators?

- Which lifestyle and demographic factors show the strongest statistical associations with diabetes?

CHAPTER 2: REVIEW OF LITERATURE

2.1 Introduction

Before undertaking original analysis, understanding the existing body of research is essential. This chapter reviews relevant literature on diabetes prevalence, lifestyle risk factors, and analytical approaches to NHANES data. The review establishes what is currently known, identifies gaps in knowledge, and provides context for the current study.

2.2 Diabetes Prevalence Trends

Long-term analysis of NHANES data reveals concerning trends in diabetes prevalence. Wang and colleagues examined data spanning from 1999 to 2018 and documented a steady increase in diagnosed diabetes among U.S. adults¹. Their study showed prevalence rising from about 10.3 percent in 1999-2000 to over 14.3 percent in 2017-2018. This upward trajectory persisted across most demographic subgroups, suggesting systemic factors driving the diabetes epidemic.

Even more concerning, the researchers found that only about 21 percent of people with diagnosed diabetes were meeting all their health goals¹. These goals include keeping blood sugar, blood pressure, and cholesterol at healthy levels. This indicates that diabetes is not only becoming more common, but many people are also struggling to manage the condition effectively.

The most recent surveillance data from the National Center for Health Statistics, covering August 2021 through August 2023, provides updated prevalence estimates². During this period, approximately 15.9 percent of U.S. adults had diabetes, with 10.8 percent having been diagnosed and an additional 5.1 percent remaining undiagnosed. The persistence of substantial undiagnosed diabetes underscores the importance of screening programs and public awareness campaigns.

2.3 Physical Activity and Sedentary Behavior

The relationship between physical activity and diabetes risk has been extensively studied. A prospective cohort study from the Multi-Ethnic Study of Atherosclerosis followed nearly 6,000 adults for over 11 years to examine how walking pace and volume related to diabetes incidence³. The researchers found that people who walked at a brisk pace had lower diabetes risk compared to those who walked slowly or did not walk at all, even after adjusting for total walking time and other confounding factors.

Importantly, this study also examined sedentary behavior measured through accelerometry. Participants who spent more time in sedentary activities, particularly television viewing, had markedly higher diabetes risk compared to those with less sedentary time³. This association remained significant even when accounting for moderate-to-vigorous physical activity levels, suggesting that prolonged sitting independently contributes to diabetes risk.

Hamilton and colleagues reviewed the physiological mechanisms underlying sedentary behavior's harmful effects⁴. When large skeletal muscle groups remain inactive for extended periods, metabolic processes that regulate blood glucose become impaired. Muscle contraction stimulates glucose uptake independent of insulin, so prolonged sitting reduces this important regulatory pathway. Additionally, sedentary time is associated with reduced lipoprotein lipase activity, contributing to metabolic dysfunction.

2.4 Obesity and Body Mass Index

The connection between excess body weight and diabetes is among the most robust findings in metabolic epidemiology. Recent NHANES data demonstrates a strong dose-response relationship, with diabetes prevalence increasing progressively across weight categories². Among adults with normal body weight, approximately 7.4 percent have diabetes. This rises to 12.2 percent among overweight individuals and reaches 24.2 percent among those with obesity.

The biological mechanisms linking obesity to diabetes involve multiple pathways. Adipose tissue, particularly visceral fat accumulated around internal organs, functions as an endocrine organ secreting inflammatory cytokines and hormones that interfere with insulin signaling. This insulin resistance forces the pancreas to produce increasing amounts of insulin to maintain normal blood glucose levels. Eventually, pancreatic beta cells become exhausted and cannot meet the elevated insulin demand, resulting in hyperglycemia and diabetes.

Body Mass Index, calculated as weight in kilograms divided by height in meters squared, serves as a practical screening tool for population studies. While BMI cannot distinguish between muscle mass and fat mass or identify fat distribution patterns, it correlates reasonably well with body fatness at the population level and predicts diabetes risk effectively in epidemiological research.

2.5 Racial and Ethnic Disparities

Diabetes prevalence varies substantially across racial and ethnic groups in the United States. The most recent national data shows that American Indian and Alaska Native adults experience the highest diabetes burden at approximately 14.5 percent, followed by Non-Hispanic Black adults and Hispanic adults both around 12.1 percent. In contrast, Non-Hispanic White adults have a prevalence of approximately 7.4 percent².

These disparities persist even after accounting for socioeconomic factors such as education and income, suggesting complex interplay of social, environmental, and possibly genetic factors. Communities of color often face structural barriers to healthy living including limited access to affordable nutritious food, fewer safe recreational spaces for physical activity, and reduced access to quality healthcare services.

Chronic stress related to experiences of discrimination and socioeconomic disadvantage may also contribute to metabolic dysregulation through neuroendocrine pathways. Cultural factors including traditional dietary practices and attitudes toward physical

activity add additional complexity to understanding and addressing these disparities.

2.6 Analytical Approaches to NHANES Data

Working with NHANES data requires understanding of survey methodology and appropriate analytical techniques. NHANES employs a complex, stratified, multistage probability sampling design to ensure results represent the non-institutionalized U.S. civilian population. This sampling approach means that standard statistical methods assuming simple random sampling may produce incorrect results if applied without modification.

For this project, I take a simplified analytical approach appropriate for an educational research project. Rather than incorporating the complex survey weights and design variables, I treat the sample as representative of the population and apply standard statistical methods⁵. This approach is acceptable for learning purposes and for identifying associations, though formal population estimates would require proper survey weighting procedures.

The R programming environment offers excellent tools for NHANES analysis. Packages such as `nhanesA` facilitate data downloading, while `dplyr` and `tidyr` enable efficient data manipulation⁵. For statistical testing, packages like `rstatix` provide user-friendly functions for common tests including Wilcoxon rank-sum tests, chi-square tests, and Kruskal-Wallis tests with appropriate post-hoc comparisons.

2.7 Gaps in Current Knowledge

Despite extensive research on diabetes epidemiology, several gaps motivate the current study. First, most published analyses use data from earlier survey cycles. The 2021-2023 cycle is the most recent complete dataset, and patterns may have evolved, particularly following the COVID-19 pandemic which significantly disrupted lifestyle behaviors for many Americans.

Second, many studies examine individual risk factors in isolation. Analyzing multiple factors simultaneously, including sedentary behavior, smoking, diet, BMI, and demographic characteristics, provides a more comprehensive picture of diabetes epidemiology. Understanding which factors show the strongest associations when considered together helps prioritize prevention efforts.

Third, while racial disparities are well-documented, continuous monitoring is necessary to assess whether public health interventions are reducing these gaps or if disparities persist or widen. The current analysis examines these patterns in the most recent available data.

Finally, from an educational perspective, there is value in replicating and extending published analyses using current data. This develops practical research skills, reinforces statistical concepts, and contributes to ongoing surveillance of this critical public health issue.

CHAPTER 3: PROBLEM FORMULATION AND RESEARCH TITLE

3.1 Justification for the Study

The need for this study emerges from several converging factors. Diabetes continues to pose an enormous public health burden with prevalence rates climbing steadily over the past two decades¹. The economic costs exceed hundreds of billions of dollars annually in the United States alone, while the human costs in terms of reduced quality of life and premature mortality are immeasurable.

Recent years have witnessed significant shifts in American lifestyles. The COVID-19 pandemic accelerated trends toward remote work and reduced physical activity. Food delivery services have proliferated, potentially affecting dietary patterns. These changes make it essential to examine diabetes risk factors using the most current data available rather than relying solely on pre-pandemic studies.

From an educational standpoint, this project demonstrates mastery of data science techniques applied to real-world health data. It showcases competence in data acquisition, preparation, statistical analysis, and interpretation. These are foundational skills for careers in health informatics, biostatistics, epidemiology, and data-driven healthcare innovation.

Additionally, examining health disparities addresses critical equity concerns². If certain populations bear disproportionate diabetes burden, identifying these patterns can inform targeted interventions and help address social determinants of health that contribute to inequalities.

3.2 Problem Statement

The central problem this research addresses is understanding the contemporary relationship between modifiable lifestyle factors and diabetes prevalence using the most recent nationally representative health surveillance data. While diabetes epidemiology has been

extensively studied, several critical gaps motivate this investigation.

First, we must establish current baseline estimates for diabetes prevalence across demographic subgroups using post-pandemic data. Second, quantifying associations between specific behavioral factors—such as daily sedentary time, smoking habits, and dietary patterns—and diabetes status requires examination in current populations. Third, comparing objective metabolic health indicators between individuals with and without diabetes provides insight into physiological differences that may inform screening protocols. Finally, identifying persistent health disparities across racial and ethnic groups remains essential for developing equitable prevention strategies.

This study addresses these needs through systematic analysis answering the seven research questions outlined in Chapter 1, Section 1.5. Rather than testing predetermined hypotheses, this exploratory investigation examines patterns and associations in observed data, establishing which factors demonstrate statistical relationships with diabetes prevalence worthy of deeper investigation in future research.

3.3 Research Title

Based on the problem statement and research objectives, the title for this project is:

Examining Lifestyle Factors and Metabolic Health Markers Associated with Diabetes Using NHANES Data (2021-2023)

This title effectively communicates several key aspects of the research. The word "Examining" indicates that this is an observational study examining associations rather than establishing causation. "Lifestyle Factors" encompasses the multiple behavioral and health-related variables being investigated, including sedentary behavior, smoking status, and dietary intake. "Metabolic Health Markers" refers to the physiological indicators such as BMI, fasting glucose, and HbA1c levels that will be analyzed. "Associated with Diabetes" clarifies that the

outcome of interest is the presence or absence of diabetes in the population. Finally, specifying "NHANES Data (2021-2023)" identifies both the data source and the specific time period, emphasizing the currency of the analysis and its relevance to contemporary health patterns.

CHAPTER 4: AIMS AND OBJECTIVES

4.1 Overall Aim

The overarching aim of this research is to systematically analyze the relationship between lifestyle factors and diabetes prevalence in the contemporary United States population using the NHANES 2021-2023 dataset. This involves describing current diabetes patterns, testing statistical associations between lifestyle variables and diabetes status, and examining health disparities across demographic groups.

4.2 Specific Objectives

To achieve the overall aim, this study has four specific objectives:

Objective 1: Data Acquisition and Preparation

The first objective is to obtain and prepare a clean, analysis-ready dataset from NHANES 2021-2023⁵. This involves downloading the appropriate data files, merging information from multiple sources, selecting relevant variables, handling missing data appropriately, and creating derived variables as needed. The final dataset must include demographics (age, gender, race/ethnicity), diabetes status, metabolic markers (BMI, fasting glucose, HbA1c), and lifestyle factors (sedentary activity minutes, smoking status, carbohydrate intake).

Objective 2: Descriptive Analysis of Diabetes Patterns

The second objective is to describe diabetes prevalence and related characteristics in the sample. This includes calculating overall diabetes prevalence, stratifying prevalence by age groups, gender, and race/ethnicity, computing summary statistics (means and standard deviations) for continuous variables in diabetes and non-diabetes groups, and creating preliminary visualizations to identify patterns in the data.

Objective 3: Statistical Testing of Associations

The third objective is to formally test whether lifestyle factors and demographic characteristics are significantly associated with diabetes status. For continuous variables, this involves conducting Wilcoxon rank-sum tests to compare distributions between diabetes and non-diabetes groups. For categorical variables, chi-square tests of independence will assess associations. All tests will use appropriate adjustments for multiple comparisons to control the false discovery rate.

Objective 4: Analysis of Racial and Ethnic Disparities

The fourth objective specifically examines health disparities across racial and ethnic groups². This involves testing whether diabetes prevalence differs significantly by race/ethnicity, comparing metabolic markers (HbA1c, fasting glucose, BMI) across racial groups using Kruskal-Wallis tests, conducting post-hoc pairwise comparisons to identify which specific groups differ, and interpreting findings in the context of health equity.

4.3 Expected Outcomes

Upon completion of these objectives, this study will produce several concrete outcomes. These include a clean, well-documented dataset ready for analysis; comprehensive descriptive statistics characterizing the sample; statistical evidence regarding which lifestyle factors show significant associations with diabetes; identification of groups experiencing health disparities; and professional visualizations communicating key findings. These outputs will form the foundation for Research Project II, where detailed results will be presented and discussed in depth.

ANNEXURES

Annexure-I: Summary of Research Project - I

Registration No: 322201210 **Name of student:** Pargat Singh

Title of the Research Project - I:

Examining Lifestyle Factors and Metabolic Health Markers Associated with Diabetes Using NHANES Data (2021-2023)

Objectives of the Research Project - I:

1. To identify appropriate NHANES 2021-2023 data files and variables for diabetes analysis.
2. To review existing literature on diabetes epidemiology and establish knowledge gaps.
3. To formulate specific research questions addressing diabetes prevalence and lifestyle associations.
4. To design analytical methodology including appropriate statistical tests.

Expected Results of the Research Project - I: Comprehensive research plan, clear problem statement, seven research questions, and methodological framework ready for implementation.

Expected outcomes of the Research Project - I: Complete literature review, identified data sources and variables, well-defined research questions, and detailed analytical methodology.

Expected Learnings from the Research Project - I: Understanding diabetes, familiarity with NHANES data structure, research question formulation skills, and statistical methodology selection.

Any Challenges/issues faced during the Research Project - I:

Navigating extensive NHANES documentation, identifying most relevant variables among hundreds available, and selecting appropriate statistical methods for different data types.

Signature of the Student: Pargat Singh

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